Roll No. | | | | | | | |

END SEMESTER EXAMINATION - 2025

Name of the Course: B. Tech Petroleum Engineering Semester: Fourth
Name of the Paper: Numerical and Geostatistical methods Paper Code: TMA-403
Time: 3 Hours Maximum Marks: 100

Note: -
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b and c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(20 marks)	CO1
(a)	Find a real root of the following equation using the secant method correct to 3 places of decimal $$f(x) = 3x^5 + 10x^4 + 3x + 10 = 0$$	
(b)	Using Newton Raphson method find the root of the equation near to x=2 correct to three places of decimal $$f(x) = x^4 - x - 10 = 0$$	
(c)	Solve the following system of linear equations using Gauss Elimination method. $$x + 4y + 9z = 16$$ $$2x + y + z = 10$$ $$3x + 2y + 3z = 18$$	
Q2	(20 marks)	CO2
(a)	Solve $\frac{dy}{dx} = x + 3y$ subject to $y(0) = 1$ choosing $h = 0.1$ and find the approximate value of y when x=1 using Euler's modified method	
(b)	Apply Explicit method to solve $\frac{dy}{dx} = x - y$ subject to $y(0) = 1$ choosing $h = 0.1$. Hence find the approximate value of y when x=0.4	
(c)	Use Runge-Kutta method of fourth order to solve: $\frac{dy}{dx} = -2xy^2$, $y(0) = 1$ and find y for x= 0.1 taking $h = 0.1$	
Q3	(20 marks)	CO3
(a)	Obtain lines of regression lines for the following data:	

x	2	4	6	8	10
y	5	7	9	8	11

(b)

x	68	64	75	50	64	80	75	40	55	64
y	62	58	68	45	81	60	68	48	50	70

Obtain the Karl Pearson's correlation coefficient for the following data:

(c) The two regression lines are $5y - 8x + 17 = 0$ *and* $2y - 5x + 14 = 0$. If $\sigma_y{}^2$=16, find,

(i) Means of x and: $\bar{x}, \bar{y}$.
(ii) Standard deviation σ_x

Q4 (20 marks) CO4

(a) Explain Geo-statistics? How it is implemented in geological work? Write its importance in geological interpretation.

(b) Explain the terms spatial analysis, data posting, contour map, Indicator map.

(c) Explain the term Kriging. How Kriging is used in estimation of values in unknown places using deterministic & probabilistic simulation technique.

Q5 (20 marks) CO5

(a) If $r_{12} = 0.8$, $r_{13} = 0.7$ and $r_{23} = 0.6$ and σ_1=10, σ_2=8 and σ_3=5, determine regression equation of X_2 on X_1 and X_3.

(b) What is Principal component analysis? Differentiate between Factor analysis and Principal component analysis. Write the applications of Principal component analysis.

(c) If $r_{12} = 0.6$, $r_{13} = 0.7$ and $r_{23} = 0.65$ and σ_1=1.0, find $S_{1.23}$.